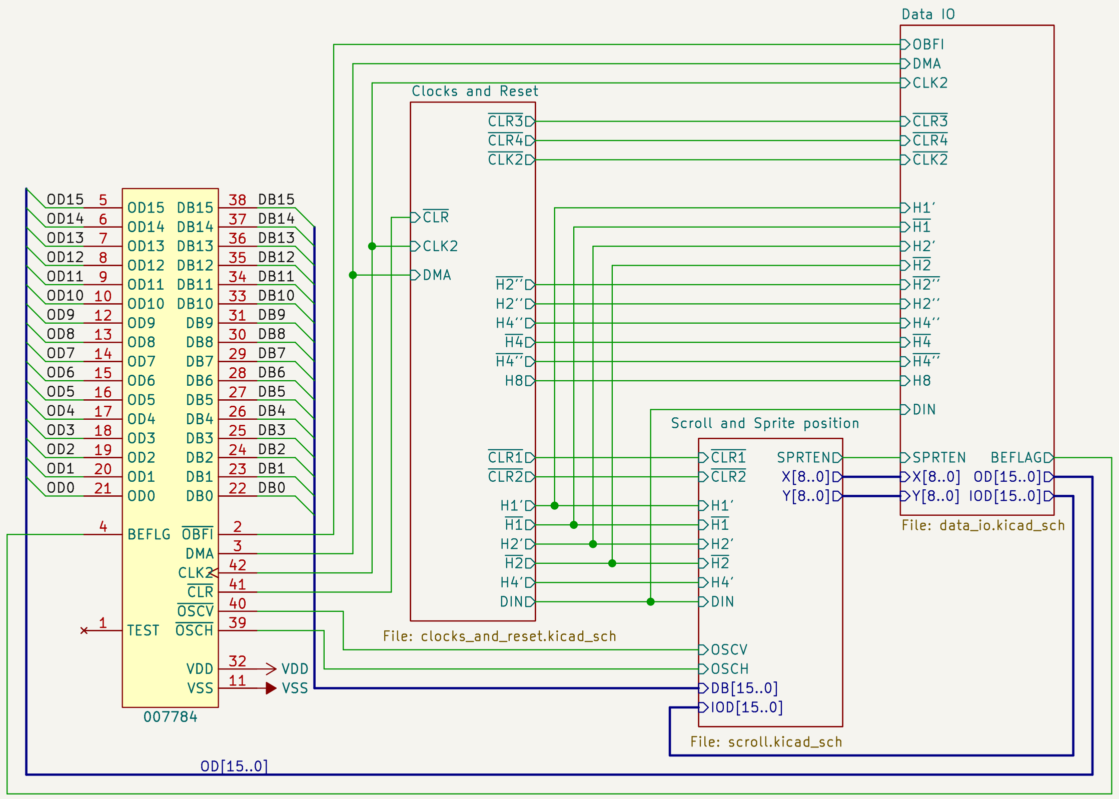
Words are parsed and written to sprite RAM in the following way:

Word 0	-> Sets/Clears BEFLAG according to bit 15.
Word 2	-> Word 3
Word 3	-> Word 0
Word 4 & 5 – XSCROLL	-> Word 1
Word 6 & 7 – YSCROLL	-> Word 2

The 007784 does the following things:

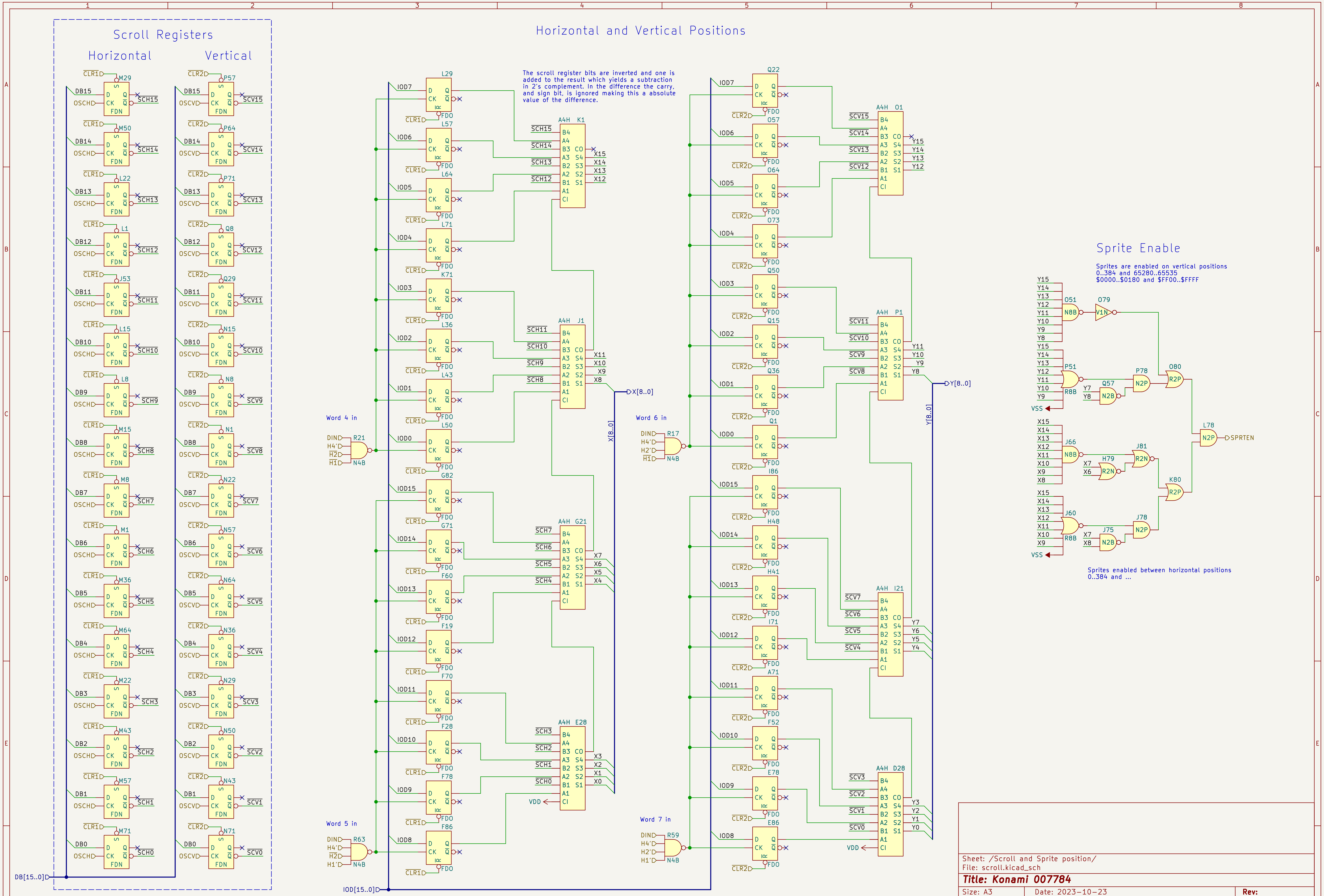
- Sets BEFLAG high if bit OD15 is set on every eight words in. Otherwise BEFLAG goes low. DMA does not need to be active.
- All other words in need DMA to be active to be parsed.
- Calculates the final vertical and horizontal positions of the sprite by using the sprite position in the unparsed object table and the horizontal and vertical scroll registers. It is written to a new parsed object table in RAM.
- The Sprite Enable (SPRTEN) bit is only set if the position is within the allowed range.
- When OBFI (OBJBUFINIT) in is high, only zeros are written to data out. That is to reset the object RAM table.



Ulf Skutnabba, twitter: @skutis77
Sheet: /
File: 007784.kicad_sch

Title: Konami 007784

Size: A3 Date: 2023-10-23 Rev:
KiCad E.D.A. 8.0.5 Id: 1/4


DB[15..0]
IOD[15..0]
X[8..0]
Y[8..0]
Word 4 in
Word 5 in
Word 6 in
Word 7 in
DIN
H4'
H2
H1
OSCH
OSCV
CLR1
CLR2
FDN
FDO
VDD
VSS
SPRTEN
M29 M50 L22 L1 J53 L15 L8 M15 M8 M1 M36 M64 M22 M43 M57 M71
P57 P64 P71 Q8 Q29 N15 N8 N1 N22 N57 N64 N36 N29 N50 N43 N71
DB15 DB14 DB13 DB12 DB11 DB10 DB9 DB8 DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0
SCH15 SCH14 SCH13 SCH12 SCH11 SCH10 SCH9 SCH8 SCH7 SCH6 SCH5 SCH4 SCH3 SCH2 SCH1 SCH0
SCV15 SCV14 SCV13 SCV12 SCV11 SCV10 SCV9 SCV8 SCV7 SCV6 SCV5 SCV4 SCV3 SCV2 SCV1 SCV0
L29 L57 L64 L71 K71 L36 L43 L50 G82 G71 F60 F19 F70 F28 F78 F86
Q22 O57 O64 O73 Q50 Q15 Q36 Q1 I86 H48 H41 I71 A71 F52 E78 E86
IOD7 IOD6 IOD5 IOD4 IOD3 IOD2 IOD1 IOD0 IOD15 IOD14 IOD13 IOD12 IOD11 IOD10 IOD9 IOD8
R21 N4B
R63 N4B
R17 N4B
R59 N4B
A4H K1
A4H J1
A4H G21
A4H E28
A4H O1
A4H P1
A4H I21
A4H D28
B4 A4 B3 A3 B2 A2 B1 A1 CI CO S4 S3 S2 S1
X15 X14 X13 X12 X11 X10 X9 X8 X7 X6 X5 X4 X3 X2 X1 X0
Y15 Y14 Y13 Y12 Y11 Y10 Y9 Y8 Y7 Y6 Y5 Y4 Y3 Y2 Y1 Y0
O51 N8B
O79 V1N
P51 R8B
Q57 N2B
P78 N2P
O80 R2P
J66 N8B
H79 R2N
J81 R2N
K80 R2P
J60 R8B
J75 N2B
J78 N2P
L78 N2P

Scroll Registers

Horizontal

Vertical

Horizontal and Vertical Positions

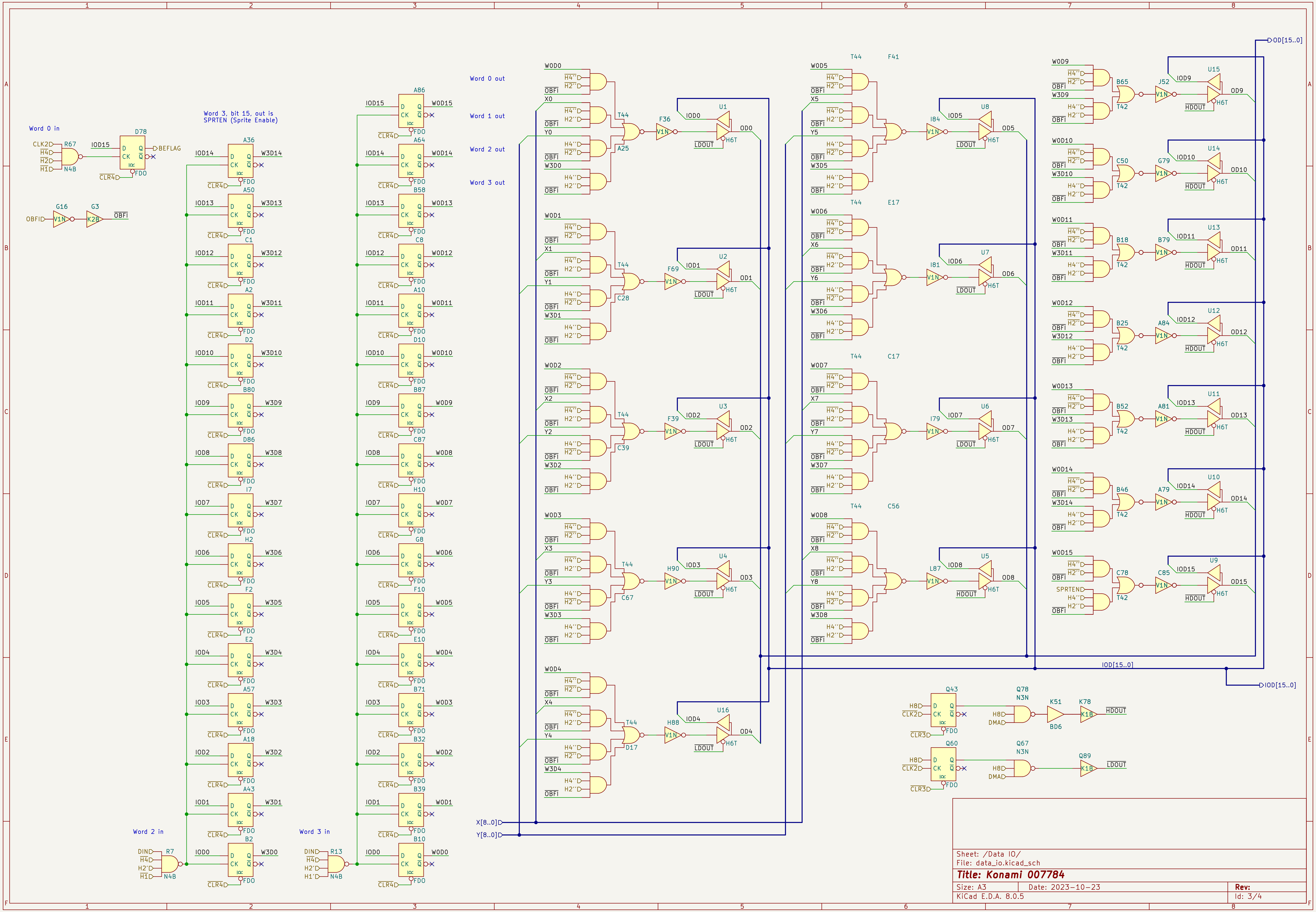
The scroll register bits are inverted and one is added to the result which yields a subtraction in 2's complement. In the difference the carry, and sign bit, is ignored making this a absolute value of the difference.

Sprite Enable

Sprites are enabled on vertical positions
0..384 and 65280..65535
$0000..$0180 and $FF00..$FFFF

Sprites enabled between horizontal positions
0..384 and ...

Sheet: /Scroll and Sprite position/
File: scroll.kicad_sch
Title: Konami 007784
Size: A3 Date: 2023-10-23 Rev:
KiCad E.D.A. 8.0.5 Id: 2/4



Sheet: /Data IO/
File: data_io.kicad_sch

Title: Konami 007784

Size: A3
Date: 2023-10-23
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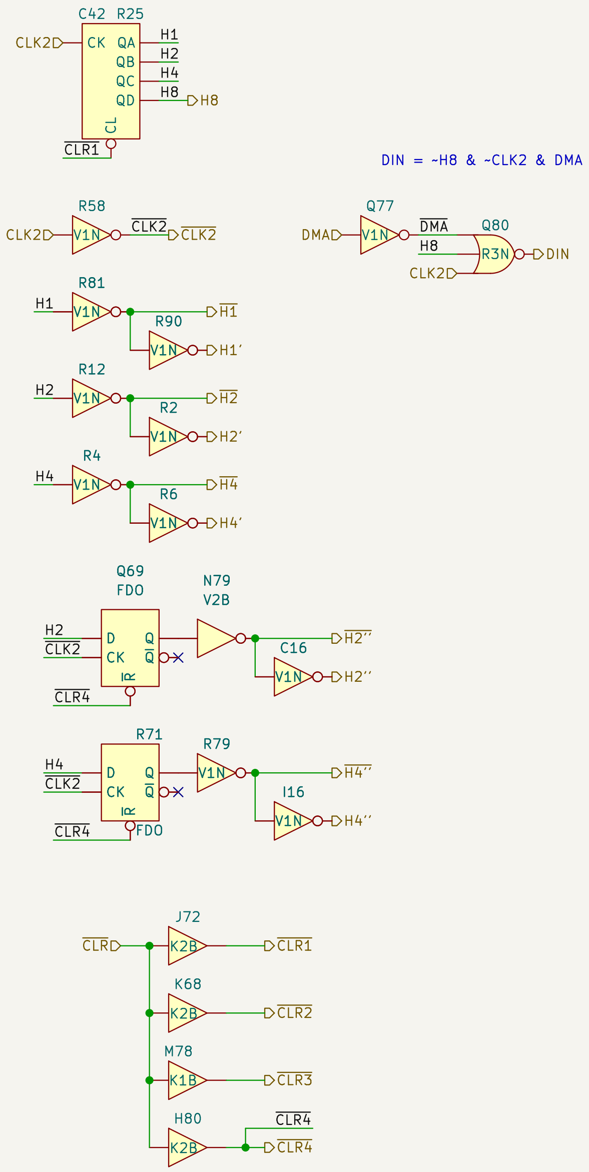
Id: 3/4


DIN = ~H8 & ~CLK2 & DMA

Sheet: /Clocks and Reset/
File: clocks_and_reset.kicad_sch

Title: Konami 007784

Size: A3
Date: 2023-10-23
Rev:
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Id: 4/4